CVOE04, ENVIRONMENTAL POLLUTION CONTROL TECHNOLOGIES

Unit I

Domestic Wastewater

1. Explain the effects of environmental pollution on human health. (05 M)
2. Describe the hydrologic cycle along with its key processes. (10 M)
3. Write the importance of hydrologic cycle. (05 M)
4. Explain the sources and types of wastewater. (10 M)
5. Describe the wastewater management within a building. (10 M)
6. Differentiate one pipe system and two pipe system of plumbing in buildings. Explain the advantages and disadvantages of one pipe system. (10 M)
7. What are the features of two pipe system? Write the advantages and disadvantages of two pipe system. (10 M)
8. Explain the wastewater treatment process with a flow diagram. (10 M)
9. Differentiate BOD and COD. Write the typical values of BOD and COD. (10 M)
10. Write the kinetic equation for BOD. Write the stages of BOD decomposition. (10 M)
11. Explain the Rotating biological contractor (RBC) type wastewater treatment system along with its key components and working principle. Also write its advantages and disadvantages. (10 M)
12. What is the working principle of Moving Bed Biological Reactor (MBBR)? Explain the key components and advantages of MBBR. (10 M)

Unit II

Industrial Wastewater

1. List the different sources of industrial wastewater. Explain the concept of 'volume reduction' in industrial wastewater treatment. (10 M)
2. Explain the different strategies to achieve concentration reduction in managing industrial wastewater. (10 M)
3. What is the importance of pre-treatment of industrial wastewater? Explain the process of 'neutralization' in industrial wastewater treatment. (10 M)
4. Explain the process 'Equalization' in Industrial Wastewater Treatment. (10 M)
5. Explain the process 'Proportioning' in Industrial Wastewater Treatment. (10 M)
6. Briefly explain the methods of Proportioning'. (05 M)

Unit III

Solid Waste Management

1. Write the classification of solid wastes based on their source of generation. (05 M)
2. Write the physical, chemical and biological characteristics of solid wastes. (10 M)
3. Explain the various stages in municipal solid waste management. (10 M)
4. What is 'composting'? Briefly explain any two methods of composting. (10 M)
5. Briefly explain the various types of landfills. (10 M)
6. Explain the key components of modern landfills. (10 M)

7. What are the impacts of landfills on environment? Briefly explain the factors influencing the landfill site selection. (10 M)
8. Explain the methodology of construction of landfill. (10 M)
9. Briefly explain the phase development and phase operation of landfills. (10 M)
10. Explain the record keeping and waste inspection at a landfill site. (10 M)
11. Explain the measures needed at the landfill site for pollution prevention and safety. (10 M)
12. Briefly explain the leachate and gas collection systems at a landfill. (10 M)
13. What are the minimum specifications for landfill cover system? (05 M)
14. Explain the environmental monitoring system at a landfill site. (10 M)
15. Explain the practical strategies of solid waste management in apartments. (10 M)

Unit IV

Air & Noise Pollution

1. Explain the history of air pollution and discuss how air pollution has evolved from ancient times to the modern industrial era.
2. Describe the various sources of air pollution and provide a detailed classification of air pollutants with suitable examples.
3. Discuss in detail the working principle, procedure, and applications of a Dust Fall Jar used in ambient air quality monitoring.
4. Explain the components, working mechanism, and operational procedure of a High Volume Air Sampler (HVAS). Add the common problems encountered during sampling.
5. "Ambient air quality monitoring is essential for environmental management." Justify this statement and explain any two monitoring instruments in detail.
6. Identify and explain the major sources of noise pollution in urban and industrial areas. How do these sources affect human health and the environment?
7. Describe the noise standards prescribed by the Ministry of Environment and Forests (MoEF) for different zones. Discuss the rationale behind these standards.
8. Explain the preventive and control measures for air pollution at (a) source level, (b) path level, and (c) receptor level.
9. Discuss various preventive measures for noise pollution. Explain both engineering and administrative controls with examples.
10. "Air and noise pollution pose significant challenges to sustainable urban development." Discuss this statement with examples and suggest suitable mitigation strategies.

Unit V

Environmental Impact Assessment

1. Define Environmental Impact Assessment (EIA). Explain the need for EIA and briefly describe EIS, FONSI, REIA, and CEIA with suitable examples.
2. Explain the step-by-step procedure involved in conducting an Environmental Impact Assessment (EIA). Highlight the importance of each stage.
3. Discuss the limitations of EIA as a planning and decision-making tool. Suggest measures to overcome these limitations.
4. Describe the major methodologies of EIA: listing, ad hoc method, network method, and overlay method. Compare their advantages and disadvantages.

5. Explain the importance of public participation in environmental decision making. Describe the objectives of public participation and explain any four commonly used public participation techniques.
6. “Public participation strengthens transparency and accountability in EIA.” Discuss this statement with suitable examples.
7. Describe the practical considerations that must be taken into account while preparing an EIA and an Environmental Impact Statement (EIS).
8. Explain the role and structure of an Environmental Impact Statement (EIS). How does FONSI differ from EIS in environmental assessment?
9. Discuss the differences between Rapid EIA (REIA) and Comprehensive EIA (CEIA). Explain the situations where each is appropriate.
10. Explain how a network method and overlay method are used in EIA. Illustrate with an example how these methods help in predicting environmental impacts.